5.3.1 Let $f \in M_k(\Gamma_0(N))$. $\forall n \in \mathbb{Z}_+$, define

$$S_n = \{d \in \mathbb{Z}^+ : d \mid n,\ \gcd(n/d, N) = 1\}$$

$$M_n = \bigcup_{d \in S_n} \bigcup_{j=0}^{d-1} \begin{pmatrix} n/d & j \\ 0 & d \end{pmatrix}$$

Show $T_n f = \sum_{\gamma \in M_n} f[\gamma]_k$.

Pf: For $n = p$, $T_p f = \begin{cases} \sum_{j=0}^{p-1} f[\begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}]_k, & p \mid N \text{ and} \\ \sum_{j=0}^{p-1} f[\begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}]_k + f[\begin{pmatrix} p & 0 \\ 0 & 1 \end{pmatrix}]_k & p \mid N \end{cases}$

$$S_p = \begin{cases} \cancel{\{1, p\},\ p \mid N} & \{p\} \quad p \mid N \\ \cancel{\{p\},\ p \nmid N} & \{1, p\} \quad p \nmid N \end{cases}$$

$$M_p = \bigcup_{d \in S_p} \bigcup_{j=0}^{d-1} \begin{pmatrix} p/d & j \\ 0 & d \end{pmatrix} = \begin{cases} \{\begin{pmatrix} p & j \\ 0 & d \end{pmatrix}, j\} \ \cancel{\cup \{\begin{pmatrix} 1 & 0 \\ 0 & d \end{pmatrix}\}} & p \mid N \\ \{\begin{pmatrix} p & j \\ 0 & d \end{pmatrix}, j\} \cup \{\begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}, j\} \end{cases}$$

For $n = p^e$, $S_{p^e} = \begin{cases} \{p^e\} & p \mid N \\ \{1, p, \cdots, p^e\} & p \nmid N \end{cases}$ ~~check for~~

For $p \mid N$, $M_{p^{e-1}} M_p = \{\begin{pmatrix} 1 & i \\ 0 & p^{e-1} \end{pmatrix}, i\}_{i=0,\cdots,p^{e-1}-1} \cdot \{\begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}, j\}_{j=0,\cdots,p-1}$

$$= \{\begin{pmatrix} 1 & j+pi \\ 0 & p^e \end{pmatrix}, \begin{matrix} i = 0, \cdots, p^{e-1}-1 \\ j = 0, \cdots, p-1 \end{matrix}\} = M_{p^e}.$$

This gives $T_{p^e} = T_{p^{e-1}} T_p$

For $p \nmid N$, $M_{p^{e-1}} M_p = \left(\{\begin{pmatrix} 1 & i_1 \\ 0 & p^{e-1} \end{pmatrix}, i_1\} \cup \{\begin{pmatrix} p & i_2 \\ 0 & p^{e-2} \end{pmatrix}, i_2\} \cup \cdots \cup \{\begin{pmatrix} p^{e-1} & i_{e-1} \\ 0 & 1 \end{pmatrix}, i_{e-1}\} \right) \cdot \{\begin{pmatrix} 1 & j \\ 0 & p \end{pmatrix}, j\}$.

$$= M_{p^e} \cup \{\begin{pmatrix} p & p(j+i_2) \\ 0 & p^{e-1} \end{pmatrix}, i_2, j\} \cup \cdots \cup \{\begin{pmatrix} p^{e-1} & p(p^{e-2}j + i_{e-1}) \\ 0 & p \end{pmatrix}, i_{e-1}, j\}$$

$$= M_{p^e} \cup \begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix} M_{p^{e-2}}.$$

This gives $T_{p^{e-1}} T_p = T_{p^e} + \begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix} T_{p^{e-2}}$.

$$f[\begin{pmatrix} p & 0 \\ 0 & p \end{pmatrix}]_k = (p^2)^{k-1} p^{-k} f = p^{k-1} \cdot \frac{1}{p} f = p^{k-1} \langle p \rangle f.$$

5.3.2 Show (5.10) $T_{p^r} = T_p T_{p^{r-1}} - p^{k-1}\langle p\rangle T_{p^{r-2}}$ $r \geq 2$

and (5.11) $T_n = \prod T_{p_i^{e_i}}$ $n = \prod p_i^{e_i}$

implies (5.12) $g(s) = \prod_p \left(1 - T_p p^{-s} + \langle p\rangle p^{k-1-2s}\right)^{-1}$.

Pf: $g(s) = \sum_{n=1}^{\infty} T_n n^{-s} = \sum_{n=1}^{\infty} \prod T_{p_i^{e_i}} p_i^{-se_i}$

$= \prod_p \sum_{r=0}^{\infty} \cancel{T_{p_i^{e_i}}} T_{p^r} p^{-rs}$.

Let $G_p(s) := \sum_{r=0}^{\infty} T_{p^r} p^{-rs}$, then

$\left(1 - T_p p^{-s} + \langle p\rangle p^{k-1-2s}\right) G_p(s) = 1$.

remark:

$\Gamma \begin{pmatrix} 1 & 0 \\ 0 & p \end{pmatrix} \Gamma \cdot \Gamma \begin{pmatrix} 1 & 0 \\ 0 & p^{r-1} \end{pmatrix} \Gamma = \Gamma \begin{pmatrix} 1 & 0 \\ 0 & p^r \end{pmatrix} \Gamma \sqcup p\, \Gamma \begin{pmatrix} p & 0 \\ 0 & p^{r-1} \end{pmatrix} \Gamma$.

5.3.3 Check Prp 5.3.1 $f \in M_k(\Gamma_1(N))$, $f(\tau) = \sum_{m=0}^{\infty} a_m(f) q^m$

$\Rightarrow T_n f(\tau) = \sum_{m=0}^{\infty} a_m(T_n f) q^m = \sum_{m=0}^{\infty} \sum_{d \mid (m,n)} d^{k-1} a_{mn/d^2}(\langle d\rangle f) \cdot q^m$.

reduces to (5.4) $a_n(T_p f) = a_{np}(f) + \chi(p) p^{k-1} a_{n/p}(f)$ $f \in M_k(N,\chi)$

if $n = p$ prime.

Pf: $f \in M_k(N,\chi) \Rightarrow a_m(T_p f) = \sum_{d \mid (m,n)} \chi(d) d^{k-1} a_{mn/d^2}(f)$

$n = p \Rightarrow d = 1$ or $p \Rightarrow a_m(T_p f) = \chi(1) \cdot 1^{k-1} a_{mp}(f) + \chi(p) \cdot p^{k-1} \cdot a_{m/p}(f)$

$= a_{mp}(f) + \chi(p) p^{k-1} a_{m/p}(f)$.

J.3.4. Show that $n=p^r$: $a_m(T_{p^r} f) = a_{mp^r}(f) + \chi(p) p^{k-1} \sum_{d \mid (m/p, p^{r-1})} \chi(d) d^{k-1} a_{mp^{r-2}/d^2}(f)$

and $n = n_1 n_2$, $(n_1, n_2) = 1$: $a_m(T_{n_1}(T_{n_2}(f))) = \sum_{d \mid (m, n_1)} \chi(d) d^{k-1} \sum_{e \mid (mn_1/d^2, n_2)} \chi(e) e^{k-1} a_{mn_1 n_2/d^2 e^2}(f)$

regrouping to (J.14) $a_m(T_n f) = \sum_{d \mid (m,n)} \chi(d) d^{k-1} a_{mn/d^2}(f)$

Pf: First equation: take out $d=1$.

$$\sum_{\substack{d \mid (m, p^r) \\ d \neq 1}} \chi(d) d^{k-1} a_{mp^r/d^2}(f) \overset{\substack{p \mid d \\ d' = d/p}}{=} \sum_{d' \mid (m/p, p^{r-1})} \chi(d') \chi(p) d'^{k-1} p^{k-1} a_{mp^{r-2}/d'^2}(f)$$

take out $\chi(p) p^{k-1}$.

Second equation: $de \mid (m, n_1 n_2)$

$\Rightarrow$ $d \mid (m, n_1)$ and $e \mid (mn_1/d^2, n_2)$.

i.e. $d \mid m$, $d \mid n_1$